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Lem unicidad series fourier

Lema 1 (Unicidad de las series de Fourier). *Sean* $f, g \in \mathcal{L}^1([0,1))$*, entonces*

$$\forall n \in \mathbb{Z} : \widehat{f}(n) = \widehat{g}(n) \implies f(x) = g(x) \text{ c.t.p.}$$

***Demostración*:** Sea $\{F_N\}_{N=1}^{\infty}$ un núcleo de sumabilidad (`lem-nucleo-fejer-nucleo-sumabilidad/`??) por el `teo-aproximacion-nucleos-sumabilidad/`??, tenemos que $\lim_{N\to\infty} \|h * F_N - h\|_1 = 0$ donde $h = f - g$.

Por otro lado, como $\widehat{f}(n) = \widehat{g}(n)$ para todo $n \in \mathbb{Z}$, tenemos que

$$h * F_N(x) = \sigma_N h(x) = \frac{1}{N}\sum_{k=0}^{N-1} S_k h(x) = \frac{1}{N}\sum_{k=0}^{N-1}\left(\sum_{n=-k}^{k} \widehat{h}(n) e^{2\pi i n x}\right) = 0.$$

Por tanto, $\lim_{N\to\infty} \|h\|_1 = 0$, luego $\|h\|_1 = 0$ y, en consecuencia, $h(x) = 0$ c.t.p., es decir, $f(x) = g(x)$ c.t.p. ■

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